

SUMMARY TABLE: ESSENTIAL MATHEMATICS GRADE 10

Sub-Strand 2.7: Volume and Capacity — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 2.7: Volume and Capacity
Driving Question	DRIVING QUESTION: Why do the tea farmers' containers — a cone, a cylinder and a bucket — look nearly the same size but hold such different amounts of water, and how can we use mathematics to design a frustum-shaped container that holds exactly the capacity we need?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 The Kericho Container Puzzle: Anchoring the Phenomenon & Launching the Driving Question	Students observed three containers used by Kericho tea farmers — a conical cover, a cylindrical drum, and a frustum-shaped bucket — that appeared nearly identical in overall size but were demonstrated (via water-fill trials or video) to hold strikingly different amounts of water. The teacher displayed all three side by side and cold-called 4–6 students to share initial predictions before any instruction. A class Driving Question Board (DQB) was started to capture student wonderings. PEDAGOGICAL NOTE: Preserve every student's 'wrong' prediction on the DQB — these become powerful revision anchors in Lessons 6–8.	Volume is the amount of three-dimensional space a solid occupies; capacity is the maximum amount of liquid a hollow container can hold. Key conversion relationships are: $1 \text{ cm}^3 = 1 \text{ mL}$, $1\,000 \text{ cm}^3 = 1 \text{ L}$, and $1 \text{ m}^3 = 1\,000 \text{ L}$. Students practised converting between units using Kenyan container examples (a 20-litre jerrican, a 500 mL cup used for porridge, a 1 m^3 water tank on a Nairobi rooftop). PEDAGOGICAL NOTE: Emphasise that $\text{cm}^3 \leftrightarrow \text{mL}$ is an exact equivalence, not an approximation — learners frequently confuse this.	Students explained why the three Kericho containers hold different amounts even though they look similar in height and width: shape determines how interior space is filled, and the mathematical formulas for each shape encode that difference. The teacher connected unit conversions back to the phenomenon by asking, 'If the cylindrical drum holds 40 L, how many cm^3 is that, and why does that number matter for a farmer choosing a container?' (Answer: $40\,000 \text{ cm}^3$; it lets the farmer compare capacities using a single unit system.) PEDAGOGICAL NOTE: Use this explanation phase to surface the idea that a formula is a compressed description of shape — a concept students will deepen through Lessons 2–5.
Lesson 2 Cone vs. Cylinder — Deriving the Cone Volume Formula	Students (or the teacher via demonstration) poured water from a conical container into a matching cylinder (same base radius r , same height h) exactly three times to fill the cylinder. This 1:3 ratio was observed repeatedly with different-sized cone–	The cone volume formula $V = \frac{1}{3}\pi r^2 h$ was derived algebraically from the observed 1:3 ratio: since $V_{\text{cylinder}} = \pi r^2 h$ and the cone fills the cylinder exactly one-third, $V_{\text{cone}} = \frac{1}{3} \times \pi r^2 h$. Students calculated the capacity of the Kericho conical cover given its radius	Students explained why the conical Kericho cover holds only one-third as much water as the cylindrical drum of the same apparent size: the cone tapers to a point, so a large portion of the cylinder's interior is simply empty space above the slanted cone

	cylinder pairs, including a model made from manila paper representing the Kericho conical tea-cover. PEDAGOGICAL NOTE: Allow 60 seconds of silent individual prediction ('How many full cones will it take?') before the pour. Cold-call at least 3 students to state their prediction with a reason — this activates prior knowledge of area and proportional reasoning.	and height, then converted the answer from cm^3 to litres. They also confirmed that for any cone, $V_{\text{cone}} = \frac{1}{3} \times V_{\text{cylinder}}$ with the same base and height. PEDAGOGICAL NOTE: Insist students write the derivation step-by-step before plugging in numbers; this prevents formula-memorisation without understanding.	wall. The teacher asked, 'What does the $\frac{1}{3}$ in the formula tell a farmer about purchasing the right container?' PEDAGOGICAL NOTE: Connect back to the DQB — students should update the board with the answer to the sub-question 'Why does the cone hold less?' Record the formula $V = \frac{1}{3}\pi r^2 h$ on the DQB anchor chart.
Lesson 3 Volume of Pyramids: Unlocking the One-Third Relationship	Students observed (via physical models or a teacher-led demonstration) that a square-based pyramid with base side a and height h fills its matching cuboid (same base, same height) exactly one-third of the way — the same ratio seen with the cone and cylinder in Lesson 2. A second observation compared a rectangular-based pyramid to its cuboid. PEDAGOGICAL NOTE: If physical models are unavailable, a carefully drawn net folded into 3D shapes works well. Ask students, 'Do you think the one-third rule is a coincidence, or a deeper mathematical truth?' — wait 30 seconds then cold-call 3 students before revealing the derivation.	The volume of any pyramid is $V = \frac{1}{3} \times \text{base area} \times \text{height}$. For a square base: $V = \frac{1}{3}a^2h$; for a rectangular base: $V = \frac{1}{3}lbh$. The same one-third relationship that links a cone to its cylinder also links a square-based pyramid to its cuboid, revealing a universal geometric principle: any pointed solid (cone or pyramid) has exactly one-third the volume of its enclosing prism or cylinder. Students practised with grain-storage pyramid models typical of Rift Valley granaries. PEDAGOGICAL NOTE: Stress the generality — the base can be any polygon; this prepares students for the frustum derivation in Lesson 5.	Students explained why grain-storage pyramidal rooftops in Rift Valley homesteads hold far less grain than a cylindrical silo of the same footprint and height: the taper removes two-thirds of the interior volume. The teacher connected this to the driving question by asking, 'How does knowing $V = \frac{1}{3} \times \text{base area} \times \text{height}$ help a Kericho farmer choose between a pointed and a flat-topped container?' PEDAGOGICAL NOTE: Update the DQB — add the generalisation 'cone and pyramid share the $\frac{1}{3}$ rule because both taper to a point.' This sets up the contrast with the frustum in Lesson 4.
Lesson 4 Introducing Frustums of Cones: Deriving the Volume Formula and Solving the Bucket Mystery	Students observed that a frustum (the bucket shape) is a cone with its top sliced off. The teacher displayed the Kericho bucket alongside a complete large cone and a small cone (the removed top), demonstrating that $V(\text{frustum}) = V(\text{large cone}) - V(\text{small cone})$. Similar-triangle diagrams were drawn on the board to show how the height H of the large cone and the height h of the removed small cone relate to the radii R and r through proportionality: $H/R = (H - h_f)/r$, where h_f is the frustum height. PEDAGOGICAL NOTE: Give students 2 minutes to attempt the subtraction setup in pairs before the teacher models it — productive struggle here is critical for ownership of the derivation.	The volume of a frustum of a cone is $V = \frac{1}{3}\pi h_f(R^2 + Rr + r^2)$, where R is the larger base radius, r is the smaller base radius, and h_f is the height of the frustum. This was derived by: (1) expressing H in terms of R , r , and h_f using similar triangles; (2) writing $V_{\text{large}} - V_{\text{small}}$ in terms of R , r , and h_f only; (3) factoring using the algebraic identity $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$. Students solved the 'bucket mystery': a Kericho farmer's frustum bucket ($R = 15 \text{ cm}$, $r = 10 \text{ cm}$, $h = 20 \text{ cm}$) holds $V = \frac{1}{3}\pi(20)(225 + 150 + 100) \approx 9\,948 \text{ cm}^3 \approx 9.95 \text{ L}$. PEDAGOGICAL NOTE: The similar-triangle step is the hardest — draw it large, label every variable, and check that all students can articulate why $H/R = (H - h_f)/r$ before moving to algebra.	Students explained the bucket mystery from the driving question: the frustum bucket holds far more than the cone ($\approx 9.95 \text{ L}$ vs. $\approx 3.3 \text{ L}$ for a comparable cone) because removing the pointed top of a cone eliminates only a small-volume tip while keeping the wide lower body. The teacher asked, 'Why does a small change in r (the top radius) produce a large change in volume?' — directing students to examine the $R^2 + Rr + r^2$ term. PEDAGOGICAL NOTE: Update the DQB — the sub-question 'Why does the bucket hold so much more than the cone?' can now be answered formally. Encourage students to re-examine their Lesson 1 predictions.
Lesson 5 Volume of Frustums of	Students observed scale models (or diagrams) of frustum-shaped grain-storage	The volume of a frustum of a square pyramid (side lengths a and b , height h) is V	Students explained how Rift Valley farmers can use $V = \frac{1}{3}h(A_1 + A_2 + \sqrt{A_1A_2})$ to

Pyramids — Designing the Perfect Grain-Storage Bin	bins used by Rift Valley farmers — truncated square pyramids with a large square base, a smaller square top, and a vertical height. The teacher demonstrated that a frustum of a square pyramid is also obtained by subtracting a small pyramid from a large pyramid, mirroring the cone-frustum derivation of Lesson 4. Students were asked to predict whether the formula would contain a term similar to $R^2 + Rr + r^2$ before the derivation. PEDAGOGICAL NOTE: Cold-call 4 students for predictions — this activates the algebraic pattern from Lesson 4 and builds transfer skills.	$= \frac{1}{3}h(a^2 + b^2 + ab)$, where a is the larger base side and b is the smaller base side. More generally, for any pyramid frustum with base areas A_1 and A_2: $V = \frac{1}{3}h(A_1 + A_2 + \sqrt{A_1A_2})$. Students used this formula to calculate volumes of grain-storage bins and compared results with the cone frustum formula, noting the structural similarity (three area-related terms summed, multiplied by $\frac{1}{3}h$). They also worked a design problem: given a required storage volume of 0.5 m^3 and a fixed base of $1 \text{ m} \times 1 \text{ m}$, find the height needed for a top side of 0.6 m. PEDAGOGICAL NOTE: The $\sqrt{A_1A_2}$ term (geometric mean of the two bases) is the conceptual highlight — spend extra time here; it surprises students that a square root appears in a volume formula.	design a grain-storage bin that holds exactly the capacity needed: by fixing two of the three variables (base area, top area, height) and solving for the third. The teacher tied this back to the driving question: 'We now have formulas for the cone, the cylinder, and both types of frustum — which shape gives a farmer the most volume for the least material?' PEDAGOGICAL NOTE: This question is intentionally left open for students to explore quantitatively in Lesson 6. Record it on the DQB as a new sub-question.
Lesson 6 Problem Solving with Real Containers: Apply & DQB Update	Students were presented with a set of real-world problem scenarios drawn from Kericho tea farming, Kisumu fish-processing, and Nairobi water-distribution contexts. Each scenario required selecting the correct formula (cone, cylinder, frustum of cone, frustum of pyramid) and performing multi-step calculations. Students first observed worked examples solved by the teacher using a think-aloud strategy, then solved parallel problems independently. PEDAGOGICAL NOTE: Use random cold-calling (lollipop sticks or a class list) for every solution step — not just the final answer — to keep all students accountable at each stage of the working.	Consolidated understanding: $V_{\text{cone}} = \frac{1}{3}\pi r^2 h$; $V_{\text{cylinder}} = \pi r^2 h$; $V_{\text{cylinder}} = 3 \times V_{\text{cone}}$ (same base and height); $V_{\text{frustum of cone}} = \frac{1}{3}\pi h(R^2 + Rr + r^2)$; $V_{\text{frustum of pyramid}} = \frac{1}{3}h(A_1 + A_2 + \sqrt{A_1A_2})$. Students practised converting final answers between cm^3, m^3, mL, and L, and interpreted answers in context (e.g., 'the frustum bucket holds 9.95 L, so the farmer needs at least 5 full buckets to fill a 50 L drum'). They also compared volumes across shapes given identical dimensions, confirming that the frustum always holds more than a cone but less than a cylinder of the same height and maximum radius. PEDAGOGICAL NOTE: Misconception to watch for — students often subtract incorrectly when applying the frustum formula; require them to show the full algebraic expansion before simplifying.	Students explained, using numerical evidence from their solved problems, exactly why the three Kericho containers 'look the same size but hold different amounts': identical external height and maximum width do not imply identical volume because the internal geometry (taper, flat base vs. pointed base) determines how much space is enclosed. The teacher facilitated a full DQB update: each original student question was revisited, and students indicated which had been answered and which remained open. PEDAGOGICAL NOTE: The DQB update is a formative assessment opportunity — photograph the board and use gaps to plan Lesson 7 group tasks.
Lesson 7 Project: Model Making & Measurement — Frustum, Pyramid Frustum, and Cone Containers	Students constructed physical scale models of a cone, a frustum of a cone, and a frustum of a square pyramid using manila paper, rulers, and scissors. They measured the dimensions of their completed models and recorded R, r, h (or a, b, h) in a data table. Groups then filled their models with sand or water (where possible) and compared the measured capacity with the	Students applied $V = \frac{1}{3}\pi r^2 h$, $V = \frac{1}{3}\pi h(R^2 + Rr + r^2)$, and $V = \frac{1}{3}h(a^2 + b^2 + ab)$ to compute predicted volumes for their hand-made models, then compared predictions with direct measurement. They identified sources of error (imprecise cutting, paper overlap, measurement rounding) and discussed how real-world engineering tolerances arise from exactly these issues	Students explained how a Kericho tea farmer (or any container manufacturer) would use the frustum formula in reverse: given a target capacity in litres, choose convenient values for two dimensions (e.g., R and h) and solve algebraically for the third (r). Each group presented their model and calculations, articulating the design trade-offs — a wider top increases capacity

	value predicted by the appropriate formula. Discrepancies between predicted and measured values were recorded and discussed. PEDAGOGICAL NOTE: Assign roles within each group (measurer, calculator, recorder, presenter) and rotate roles mid-task to ensure all students engage with both the practical and mathematical components.	— connecting to the driving question about designing a frustum container that holds exactly a required capacity. PEDAGOGICAL NOTE: Require each group to calculate the percentage error between predicted and measured volume; this reinforces both the formula and practical measurement skills within a single activity.	but uses more material. The teacher facilitated peer feedback focused on mathematical accuracy and clarity of explanation. PEDAGOGICAL NOTE: Insist that presentations include the formula, substitution, and unit-converted answer — not just the final number. This prepares students for the Lesson 8 assessment.
Lesson 8 Model Building Culmination, Final Explanation & Assessment: The Kericho Farmers' Containers Solved	Students revisited the original Lesson 1 anchoring phenomenon: the three Kericho containers side by side. The teacher re-showed (or recalled) the water-fill demonstration from Lesson 1 and asked students to now explain every observation using mathematical evidence from Lessons 2–7. The class reviewed their Lesson 1 predictions on the DQB, and each student individually annotated a diagram of all three containers with the correct formula and a calculated volume using given dimensions. PEDAGOGICAL NOTE: The deliberate return to the Lesson 1 phenomenon is essential for closure — students should feel the intellectual satisfaction of having fully explained a puzzle that once seemed mysterious. Allow 3 minutes of silent individual reflection before whole-class discussion.	Full synthesis: (1) A cone and a cylinder with the same base radius and height always satisfy $V_{\text{cylinder}} = 3 \times V_{\text{cone}}$, which is why the Kericho cylindrical drum holds three times as much as the matching conical cover despite looking similar. (2) The frustum bucket holds significantly more than the cone because the frustum retains the wide lower body of the cylinder while removing only the small-volume pointed tip; this is captured by the formula $V = \frac{1}{3}\pi h(R^2 + Rr + r^2)$. (3) Any frustum-shaped container (cone or pyramid) can be designed to hold exactly a specified capacity by fixing two dimensions and solving for the third using the appropriate formula. (4) Unit conversions ($1 \text{ cm}^3 = 1 \text{ mL}$; $1\,000 \text{ cm}^3 = 1 \text{ L}$; $1 \text{ m}^3 = 1\,000 \text{ L}$) are the bridge between mathematical volume and practical capacity. PEDAGOGICAL NOTE: Use this lesson's assessment task as a summative check — a two-part question asking students to (a) calculate and compare volumes of given cone, cylinder, and frustum containers, and (b) design a frustum with a specified target capacity, showing all working.	Students produced a final written explanation answering the driving question in full: the Kericho farmers' containers look similar because they share approximately the same height and maximum diameter, but the cone holds only $\frac{1}{3}$ as much as the cylinder (because of the $\frac{1}{3}$ factor in $V = \frac{1}{3}\pi r^2 h$), while the frustum bucket holds an intermediate amount determined by $V = \frac{1}{3}\pi h(R^2 + Rr + r^2)$. A farmer can use this mathematics to design a frustum container of any desired capacity by selecting R, r, and h to satisfy the equation. The teacher conducted a whole-class DQB close-out: every sub-question was marked as 'answered' with the lesson number and key formula noted beside it. PEDAGOGICAL NOTE: Collect written explanations as a portfolio artefact. Strong responses will cite specific formulas, substitute real numbers, convert units, and connect the mathematical result back to the farmer's practical decision — all four elements should appear in the marking rubric.

ATTRIBUTION AND LICENSING

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